HOSPITAL MANAGEMENT SYSTEM

CODE COVERAGE REPORT

Software Engineering Project · LLVM source-based coverage · 6 November 2026

PRIMARY RESULT	Line coverage: 86.67% | 2,691 of 3,105 production lines

86.67%	90.20%	97.92%	81.98%
Line coverage	Region coverage	Function coverage	Branch coverage

1. Purpose and scope

This report evaluates how much of the Hospital Management System's production source code was exercised during the recorded automated test run. LLVM source-based instrumentation was used to measure line, region, function, and branch coverage. The report is restricted to the production executable's coverage mapping; test-source paths are excluded.

2. Aggregate coverage results

Metric	Covered	Missed	Total	Coverage
Lines	2,691	414	3,105	86.67%
Regions	1,638	178	1,816	90.20%
Functions	188	4	192	97.92%
Branches	1,037	228	1,265	81.98%

3. Production source-file breakdown

Production source	Line coverage	Branch coverage
Appointment.cpp	83.68%	88.69%
Doctor.cpp	97.24%	95.45%
FileManager.cpp	77.65%	74.13%
HospitalSystem.cpp	90.31%	80.59%
Patient.cpp	97.80%	97.06%
main.cpp	100.00%	N/A

4. Analysis and findings

Strongly covered components. Patient.cpp (97.80% line coverage) and Doctor.cpp (97.24%) show high execution coverage. The main.cpp entry point is fully covered for its six instrumented lines. Appointment.cpp records 83.68% line coverage.

Primary improvement area. FileManager.cpp has the lowest line coverage (77.65%) and branch coverage (74.13%). Additional tests should focus on CSV parsing and escaping, persistence failures, invalid or missing files, and path/symlink validation. HospitalSystem.cpp is the next major area for targeted tests, particularly decision paths around workflows and validation.

Interpretation. The 86.67% result means 2,691 of 3,105 instrumented production lines were executed in the measured run. The 414 missed lines identify code not observed in that run; they are not automatically defects, but should be reviewed for relevance and risk.

5. Measurement method

The production executable build-llvm/hms.exe was analyzed with llvm-cov using the merged LLVM profile build-llvm/coverage.profdata. Test-source paths were filtered with the tests-directory exclusion. The production-only report completed without an LLVM mismatched-data warning. The test executables compile FileManager.cpp with HMS_TESTING conditionals, so their coverage was not merged into this production-only result.

6. Limitations

Coverage is an execution metric, not a correctness metric. It does not establish that every assertion, requirement, input combination, or real-world workflow is correct. Line and branch percentages should be interpreted alongside the test specifications, expected results, defect records, and manual system validation. A high percentage does not guarantee the absence of defects.

7. Evidence and reproducibility

Evidence artifact	Purpose
build-llvm/production-coverage-final.txt	Human-readable production coverage summary
build-llvm/production-coverage-final.json	Machine-readable LLVM coverage export
build-llvm/ctest-results.xml	JUnit-format test execution record
build-llvm/coverage.profdata	Merged LLVM instrumentation profile

Execution context: 51 CTest tests passed, 0 failed, in 31.47 seconds. Source revision: 542b90be8adaddf13f7ce900340973eb4c9f5a56.